XIANLONG MAI

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References

- Prof. Shuaishuai Sun (USTC)   PhD supervisor   Email: sssun@ustc.edu.cn   Mobile phone: +8618119643257
- Prof. Antonio Frisoli (SSSA)   Co-supervisor   Email: antonio.frisoli@santannapisa.it   Mobile phone: +39050882549
- Prof. Guolin Yun (USTC)   Collaborator   Email: ygl@ustc.edu.cn   Mobile phone: +8613695516039

Education

Visiting PhD student — Nov 2024 - Apr 2026
Haptics and Teleoperation (Supervised by Prof. Antonio Frisoli) — Pisa, Italy
Sant'Anna School of Advanced Studies (SSSA)

PhD candidate — Sep 2021 - Jun 2026
Instruments Science and Technology (Supervised by Prof. Shuaishuai Sun) — Hefei, Anhui, China
University of Science and Technology of China (USTC)

Bachelor of Science — Sep 2017 - Jun 2021
Theoretical and Applied Mechanics — Hefei, Anhui, China
University of Science and Technology of China (USTC)

Research Interests

- Haptics in teleoperation
- Assistive exoskeletons
- Smart materials and their applications in wearable devices

Main Research Experiences

1. Nonmotorized Hand Exoskeleton for Rescue and Beyond: Substantially Elevating Grip Endurance and Strength (09/2021 - 06/2024)

- Description: Design of a magnetorheological-grease-based hand exoskeleton system that can largely elevate grip endurance with low energy and enhance grip strength by energy harvesting on finger movement.
- Key roles: 1) Structural design, simulation and modeling. 2) Circuit development, signal processing and system integration. 3) Experiments design and results analysis. 4) Paper writing and figure drawing.

2. A Compact and Wide-Force-Range Hand Exoskeleton for Whole-Hand Kinesthetic Feedback and Tracking in Teleoperation (07/2024 - 07/2025)

- Description: Integrating a hand exoskeleton with a compact magnetorheological (MR) brake and potentiometer for kinesthetic feedback in teleoperation. Characterization and experiments show a wide force range in kinesthetic feedback and effectiveness in creating immersive sensation and reducing object handling force in teleoperation.
- Key roles: 1) Structural design, simulation and modeling. 2) Circuit development, signal processing and teleoperation system integration. 3) Experiments design and results analysis. 4) Paper writing and figure drawing.

3. A hand exoskeleton with hybrid active-passive actuation for kinesthetic feedback in virtual reality and teleoperation (08/2025 - present)

- Description: Developing a new prototype of kinesthetic feedback hand exoskeleton for virtual reality and teleoperation, utilizing the advantages of MR brake (passive, wide-force-range) and brushless DC motor (active, fast response).

- Key roles: 1) Structural design, simulation and modeling. 2) Algorithm design for hybrid actuation.

Publications

First-author papers

- **X. Mai** et al., "Nonmotorized Hand Exoskeleton for Rescue and Beyond: Substantially Elevating Grip Endurance and Strength," ***IEEE Transactions on Robotics (JCR Q1, IF 10.5)***, vol. 41, pp. 4761-4775, 2025, doi: 10.1109/TRO.2025.3588750.
- **X. Mai** et al., "A Compact and Wide-Force-Range Hand Exoskeleton for Whole-Hand Kinesthetic Feedback and Tracking in Teleoperation," ***IEEE/ASME Transactions on Mechatronics*** **(under review)**, 2025.

Co-author papers

- W. Li, **X. Mai** et al., "Development of a magnetorheological hand exoskeleton featuring a high force-to-power ratio for enhanced grip endurance," ***Smart Materials and Structures （JCR Q1， IF 3.8）***, 2025, doi: 10.1088/1361-665X/ae1bf0.
- L. Li, **X. Mai** et al., "Development of upper limb exoskeleton with high braking torque density and low power consumption," ***Journal of Intelligent Material Systems and Structures （JCR Q3, IF 2.5）***, 2025, doi: 10.1177/1045389X251390002.
- R. Chen, **X. Mai** et al., "A Soft Fabric-Based Thermal Haptic Device for Virtual Reality," ***IEEE Access (JCR Q2, IF 3.7)***, 2026.
- E.A.T. Yew, **X. Mai** et al., "Design and Development of a Unified Dual Electric and Magnetorheological Robotic Joint," ***IEEE/ASME Transactions on Mechatronics*** **(under review)**, 2025.
- Y. Li, H. Wang, J. Yang, **X. Mai** et al., "An MRE-based Vacuum Adhesion Device Capable of Working on Rough Surfaces with Large Adhesion Force for Climbing Robots," ***IEEE/ASME Transactions on Mechatronics*** **(under review)**, 2025.

Patent

- Shuaishuai Sun, **Xianlong Mai** et al., "Exoskeleton," Invention patent, CN114536379A, Authorized. (2022)

Research Skills

Experimental skills:

- Development experience with STM32 and ESP32 microcontroller boards.
- Familiar with sensor data acquisition and collaborative control (e.g., force/torque/distance/hall sensors).
- Experience with 3D printing, material testing system (MTS) and oscilloscope.
- Experience with power amplifier, motor driver board, and ADC/DAC.

Coding and software skills:

- Programming: C/C++, MATLAB, LABVIEW
- 3D modeling/CAD: SOLIDWORKS, UG, Auto CAD
- Simulation: COMSOL Multiphysics
- Data analysis and visualization: ORIGIN
- Illustration and video editing: Adobe Illustrator, Keyshot, Adobe Premiere

Languages: Mandarin (native), English (C1)

Honors & Awards

USTC 'Six Excellence Student of the Year' (1 of 10 awardees)	2022
Graduate Academic Scholarship	2021, 2022, 2023, 2024, 2025
Outstanding Student Scholarship (Grade 3)	2018, 2019, 2020
Finalist in the USTC “Robogame” Robotics Competition	2019
Second Prize for Anhui division in Zhou Peiyuan National University Student Mechanics Competition	2019

Teaching Experiences

Spring 2023   Teaching Assistant   Academic English Writing   USTC

Spring 2024   Teaching Assistant   Academic English Writing   USTC

Grant Applications

- Student’s Innovation and Entrepreneurship Foundation of USTC - "A Hand Enhancement Exoskeleton Design Based on Magnetorheological Technology"(CY2022G01) - Project leader - 2022.11-2023.10
- Student’s Innovation and Entrepreneurship Foundation of USTC - "Upper limb exoskeleton based on high-efficiency, energy-saving, and compact actuator"(CY2023G011) - Participated - 2023.11-2024.10
- Student’s Innovation and Entrepreneurship Foundation of USTC - "A multimodal quadruped robot based on magnetorheological technology"(CY2023G006) - Participated - 2023.11-2024.10

Services

Journal reviewer:

- IEEE Robotics and Automation Letters
- Smart Materials and Structures

Conference reviewer:

- 19th IEEE/RAS-EMBS International Conference on Rehabilitation Robotics (ICORR 2025)
- EuroHaptics 2026